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Accredited **Grade 'A'** by NAAC



QS 5 Star Rating for Academic Development,
Employability, Facilities and Program Strength



Perfect score of **150/150** as a testament
to exceptional E-Learning methods

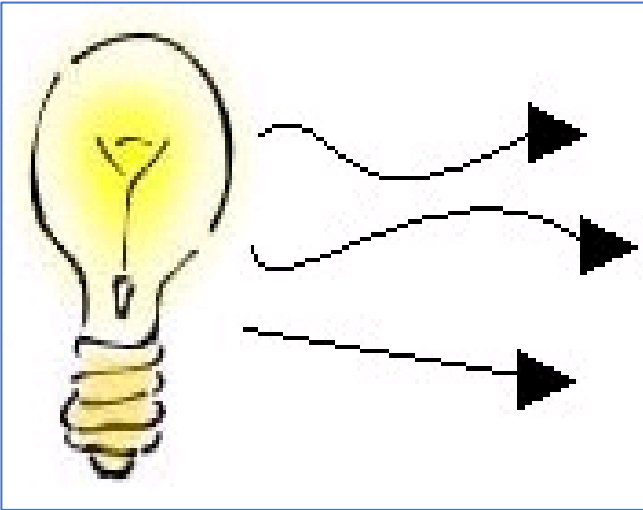
Lasers

(Light Amplification by Stimulated Emission of Radiation)

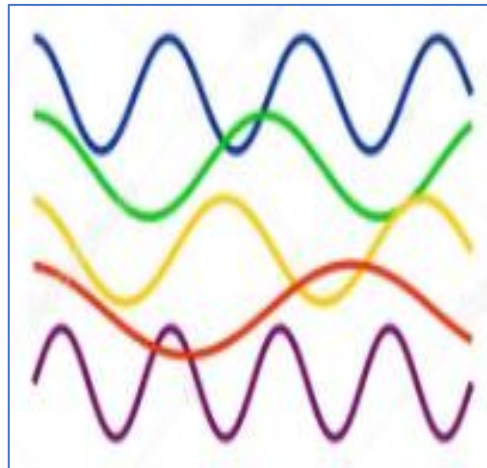
Outline

- Introduction
- Laser – Properties & Characteristics
- Absorption, Spontaneous & Stimulated Emissions of Radiation
- Einstein's Coefficients
- Population Inversion & Pumping
- Basic Requirements
- Ruby Laser – Construction & Working
- Helium Neon laser – Construction & Working
- Applications

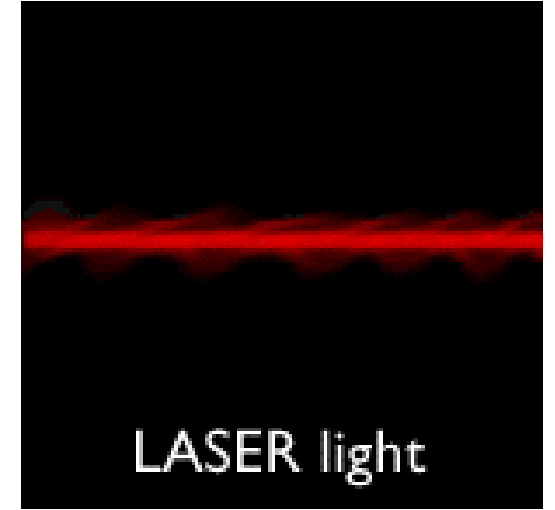
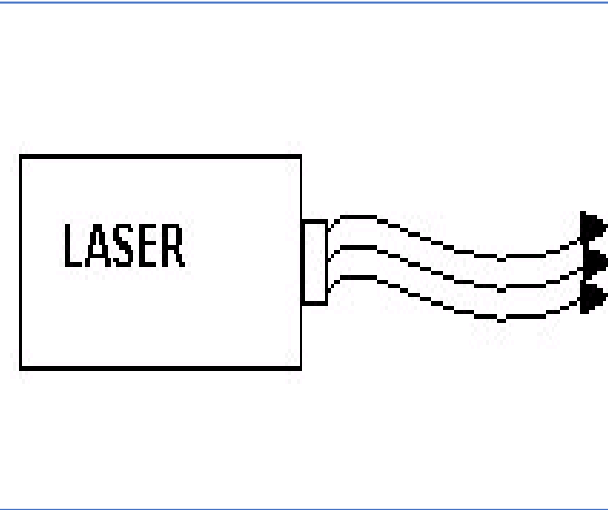
Incandescent Light



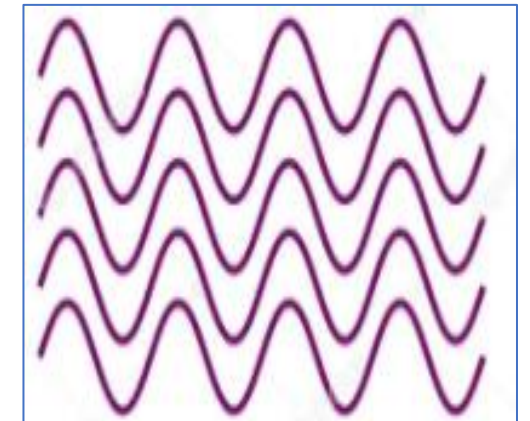
- Many wavelengths
- Omnidirectional
- Incoherent
- Relatively weak Intensified
- Spot Focus – Not Possible
- Irradiance Independent on direction



Laser Light



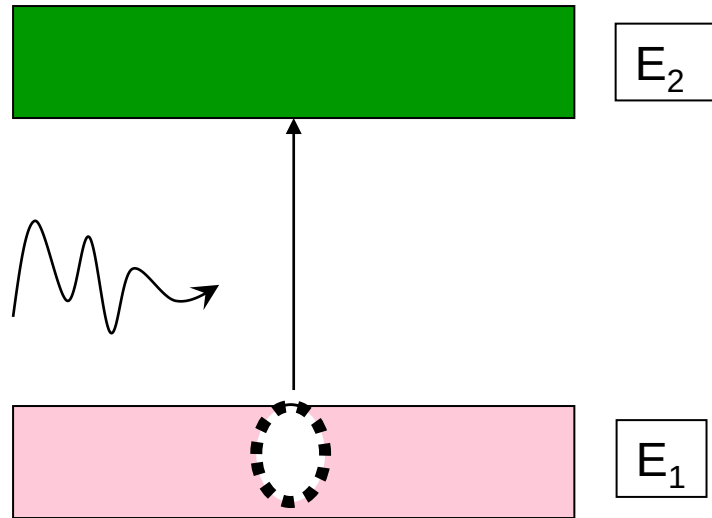
- Monochromatic
- Unidirectional
- Highly Coherent
- Intensified beam (dark)
- Spot Focus – Possible
- Irradiance dependent on direction



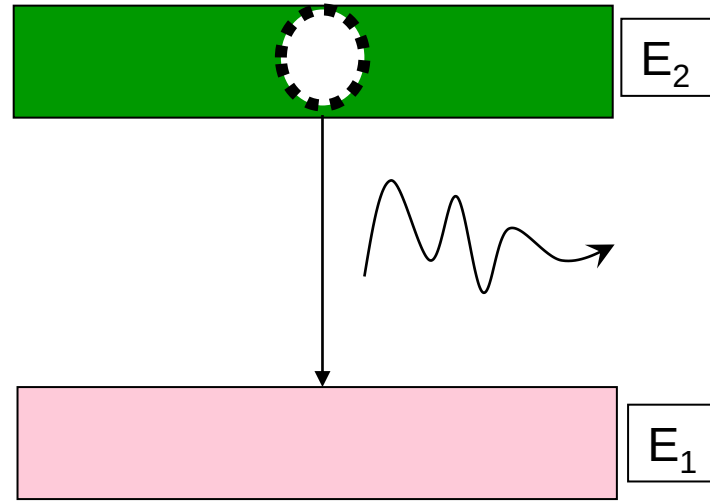
History of Lasers – Properties & Characteristics

- Einstein – 1917, Charles Townes – 1954 (MASER), Prochosov & Basov – 1954 (MASER)
- In 1958, Charles Townes and Arthur Schawlow proposed concept of Laser.
- Townes, Basov and Schawlow - 1964 - Nobel prize - LASER - MASER principle
- Theodore Maiman - 1960 - Ruby Laser, $\lambda = 6943 \text{ \AA}$
- Ali Jawan - 1961- Helium Neon Laser, $\lambda = 6328 \text{ \AA}$
- Acronym - **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation
- A device produces a coherent beam of optical radiation by stimulating electronic, ionic, or molecular transitions to higher energy levels
- Monochromatic/ Coherence/ Coherence length
Sodium Vapor lamp - 0.3 mm and He – Ne laser - 100 m
- High Power – power - 10^5 watt , - Cutting and Welding
- Focusing - High Directionality – spot of area $(\mu\text{m})^2$ - surgery, CDs, material processing...
- Spectral Purity – spectral width - $\Delta\lambda = 10^{-6} \text{ \AA}$ - Holography, Optical communication and spectroscopy
- Divergence, $\phi = (d_2 - d_1)/(l_2 - l_1)$, here d and l are diameter of spot at a specific distance
Ordinary light - Spherical wave fronts – Highly Divergent
- Laser – Plane Waves, Less divergent

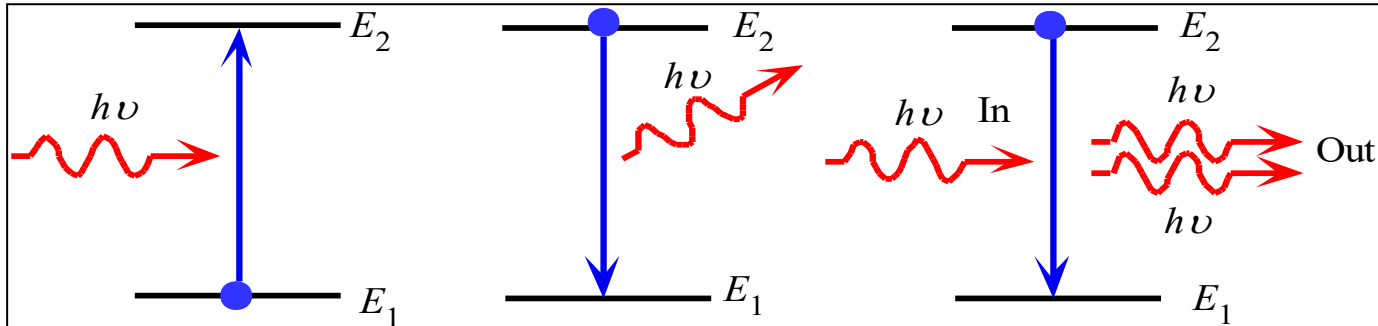
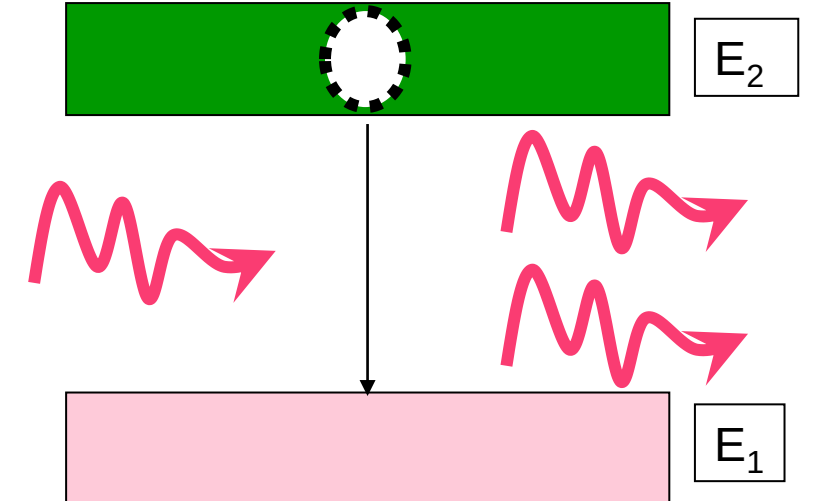
Stimulated Absorption



Spontaneous Emission



Stimulated Emission



(a) Absorption (b) Spontaneous emission (c) Stimulated emission

Absorption, spontaneous (random photon) emission and stimulated emission.

Spontaneous Emission

- Transition from Higher to lower energy level
- Incident Photon is not required
- Single Photon is emitted
- Energy of emitted photon – difference of energy levels
- Incoherent Radiation
- Minimum Intensity
- Less directionality and more angular spread
- Constant Probability function
- Bohr

- Example: Sodium vapor lamp or mercury vapor lamp

Stimulated Emission

- Transition from Higher to lower energy level
- Incident Photon of energy equal to the difference of two energy levels is required
- Two coherent Photons are emitted
- Energy of emitted photon is twice the energy of the incident photon
- Coherent Radiation
- Intensity is maximum
- High directionality and less angular spread
- Variable probability factor
- Einstein
- Example: Light from laser source

Necessary Condition for the laser operation population inversion

If $N_1 > N_2$

- radiation is mostly absorbed
- spontaneous radiation dominates.

if $N_2 \gg N_1$ - population inversion

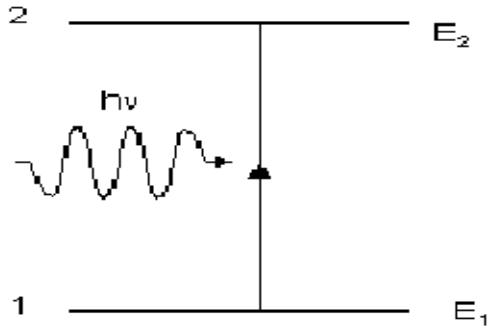
- most atoms occupy level E_2 , weak absorption stimulated emission prevails
- light is amplified

Population Inversion

- This situation in which the number of atoms in the higher state exceed that in the lower state ($N_2 > N_1$) is known as population inversion.

Einstein's Coefficients

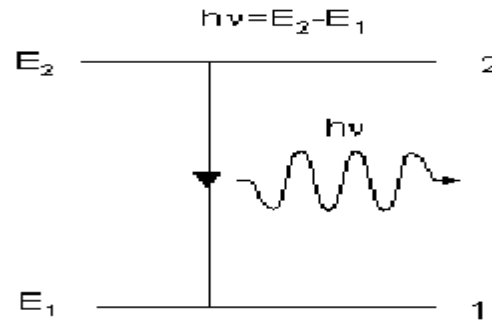
Stimulated Absorption



$$h\nu + E_1 = E_2$$

$$P_{12} = B_{12}u(\nu)$$

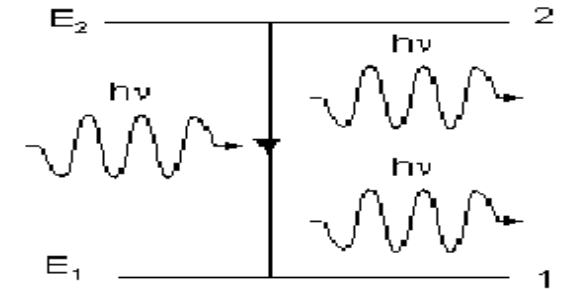
Spontaneous Emission



$$E_2 - E_1 = h\nu$$

$$P_{21}^I = A_{21}$$

Stimulated Emission



$$P_{21} = A_{21} + B_{21}u(\nu)$$

B_{12} - Einstein's Coefficient of Stimulated Absorption

A_{21} - Einstein's Coefficient of Spontaneous Emission

B_{21} - Einstein's Coefficient of Stimulated Emission

Relation between the Einstein's Coefficients

Let N_1 and N_2 be the number of particles (population) in the ground and the excited states, having energies of E_1 and E_2 respectively.

The total probability rate of excitation due to absorption $N_1 P_{12} = N_1 B_{12} u(\nu)$

The total probability rate of de-excitation due to emission $N_2 P_{21} = N_2 [A_{21} + B_{21} u(\nu)]$

At thermal equilibrium, the probability rates of absorption and emission are equal, given by

$$\begin{aligned} N_1 P_{12} &= N_2 P_{21} \\ \Rightarrow N_1 B_{12} u(\nu) &= N_2 [A_{21} + B_{21} u(\nu)] \\ &= N_2 A_{21} + N_2 B_{21} u(\nu) \end{aligned}$$

(or)

$$\begin{aligned} N_2 A_{21} &= N_1 B_{12} u(\nu) - N_2 B_{21} u(\nu) \\ &= u(\nu) [N_1 B_{12} - N_2 B_{21}] \\ \Rightarrow u(\nu) &= \frac{N_2 A_{21}}{[N_1 B_{12} - N_2 B_{21}]} \end{aligned}$$

Relation between the Einstein's Coefficients

$$\begin{aligned}
 u(\nu) &= \frac{N_2 A_{21}}{[N_1 B_{12} - N_2 B_{21}]} \\
 &= \frac{A_{21}}{\left[\left(\frac{N_1}{N_2}\right) B_{12} - B_{21}\right]} \\
 &= \frac{A_{21}}{B_{21} \left[\left(\frac{N_1}{N_2}\right) \left(\frac{B_{12}}{B_{21}}\right) - 1\right]} \\
 &= \left[\frac{A_{21}}{B_{21}}\right] \frac{1}{\left[\left(\frac{N_1}{N_2}\right) - 1\right]} \quad (\because B_{12} = B_{21}) \\
 \therefore u(\nu) &= \left[\frac{A_{21}}{B_{21}}\right] \frac{1}{\left[\exp\left(\frac{h\nu}{KT}\right) - 1\right]}
 \end{aligned}$$

From Boltzmann distribution, we have, $N_r = N_0 \left[\exp\left(\frac{-E_r}{KT}\right) \right]$

Number of particles in the first state, $N_1 = N_0 \left[\exp\left(\frac{-E_1}{KT}\right) \right]$

Number of particles in the second state, $N_2 = N_0 \left[\exp\left(\frac{-E_2}{KT}\right) \right]$

$$\left(\frac{N_1}{N_2}\right) = \exp\left(\frac{E_2 - E_1}{KT}\right) = \exp\left(\frac{h\nu}{KT}\right)$$

Relation between the Einstein's Coefficients

From the Planck's Radiation Law, the energy density is given by

$$u(\nu) = \left[\frac{8\pi h \nu^3}{c^3} \right] \left[\frac{1}{\exp\left(\frac{h\nu}{KT}\right) - 1} \right]$$

Comparing the energy density, $u(\nu) = \left[\frac{A_{21}}{B_{21}} \right] \frac{1}{\left[\exp\left(\frac{h\nu}{KT}\right) - 1 \right]}$, with Planck's law, we get

$$\frac{A_{21}}{B_{21}} = \left(\frac{8\pi h \nu^3}{c^3} \right)$$

Inference: The ratio between the Einstein's coefficient of Spontaneous to stimulated emission is directly proportional to the cube of the frequency of the incident photon.

Pumping: Population Inversion can be achieved by exciting the medium (atoms) from their ground state to an excited state with suitable form of energy. This process is called pumping. The objective is to obtain a non-thermal equilibrium.

There are several methods of pumping a laser and producing population inversion necessary for the occurrence of stimulated emission.

Optical Pumping: If the luminous energy (intensity) is supplied to the medium (atoms) for causing population inversion, then the pumping is called optical pumping. (The atoms are excited by bombarding them with photons.)

In optical pumping, the luminous energy is usually come from a light source in the form is short flashes of light. Thus method was first used in Ruby Laser by Maiman and now a days used in solid state lasers.

Example: Ruby Laser (Lasers that have a short-lived population inversion produce pulsed output – these are **pulsed** lasers).

Electrical Pumping: The pumping by electric discharge is preferred in gaseous ion lasers (He-Ne Laser).

In discharge tube, when an potential difference is applied b/w cathode and anode, the electrons emitted from cathode are accelerated towards the anode. Some of these electrons collides with the atoms of the active medium, ionize the medium (atoms) and raise these to the higher energy levels. This produces the required population inversion.

This is also called the **Direct Electron Conversion**.

So, the atoms are excited by Electron collision in a discharge tube.

Lasers that maintain a population inversion indefinitely produce continuous output – termed CW (for continuous wave) lasers.

In-elastic atom atom collision: In electric discharge, one type of atoms are raised to their excited states. The excited atoms collide in-elastically with another type of atoms. These another type of atoms provide the population inversion needed for laser action. This pumping is also used in He-Ne Laser.

Direct Conversion: A direct conversion of electric energy in to radiant energy occurs in the light emitting diodes (LEDs). This process of pumping is used in semiconductor diodes.

Chemical Conversion: In a chemical conversion, energy comes from a chemical reaction with any need of the other energy sources.

Eg. Hydrogen can combine with Fluorine to form hydrogen-fluoride.



This reaction is used to pump a CO₂ laser.

Population Inversion

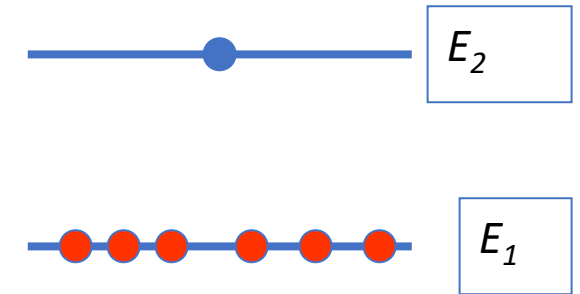
From Boltzmann distribution, we have, $N_r = N_0 \left[\exp \left(\frac{-E_r}{KT} \right) \right]$

Number of particles in the first state, $N_1 = N_0 \left[\exp \left(\frac{-E_1}{KT} \right) \right]$

Number of particles in the second state, $N_2 = N_0 \left[\exp \left(\frac{-E_2}{KT} \right) \right]$

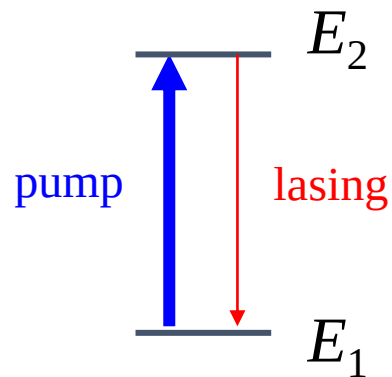
- At thermal equilibrium, $N_1 > N_2$, $E_1 < E_2$
- This situation in which the number of atoms in the higher state exceed that in the lower state ($N_2 > N_1$) is known as population inversion.
- Population Inversion (or) Inverted Popular state - Making $N_2 > N_1$

$$\left(\frac{N_1}{N_2} \right) = \exp \left(\frac{E_2 - E_1}{KT} \right) = \exp \left(\frac{h\nu}{KT} \right) \quad \left(\frac{N_2}{N_1} \right) = \exp \left(-\frac{h\nu}{KT} \right)$$
- Negative Temperature state (non-equilibrium/ unstable state)

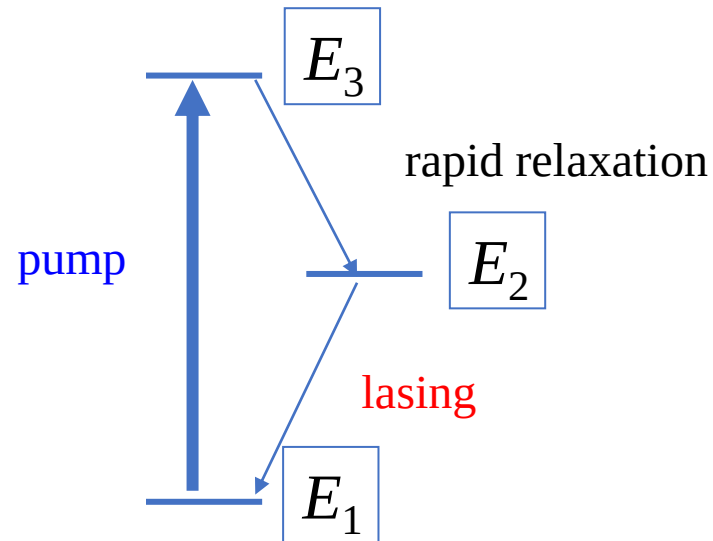


Process of producing population inversion

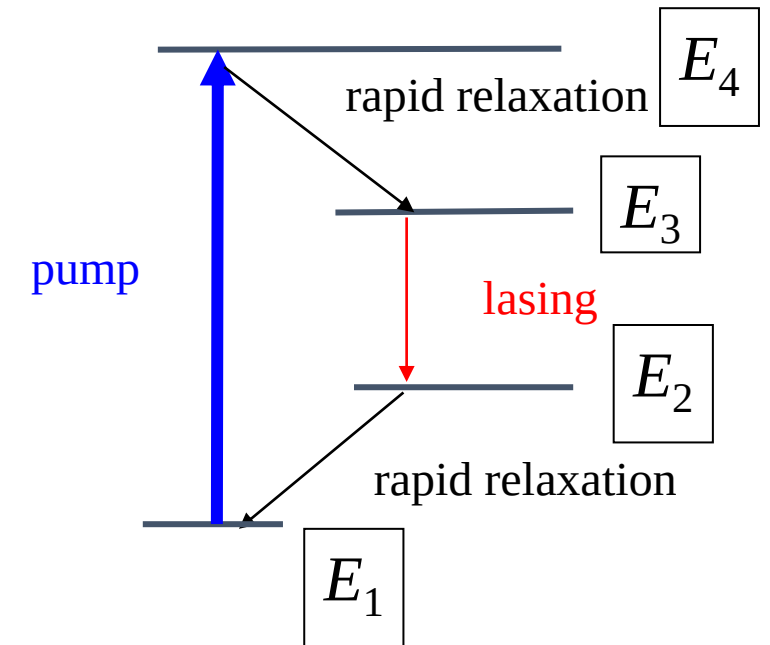
Two-level system



Three-level system

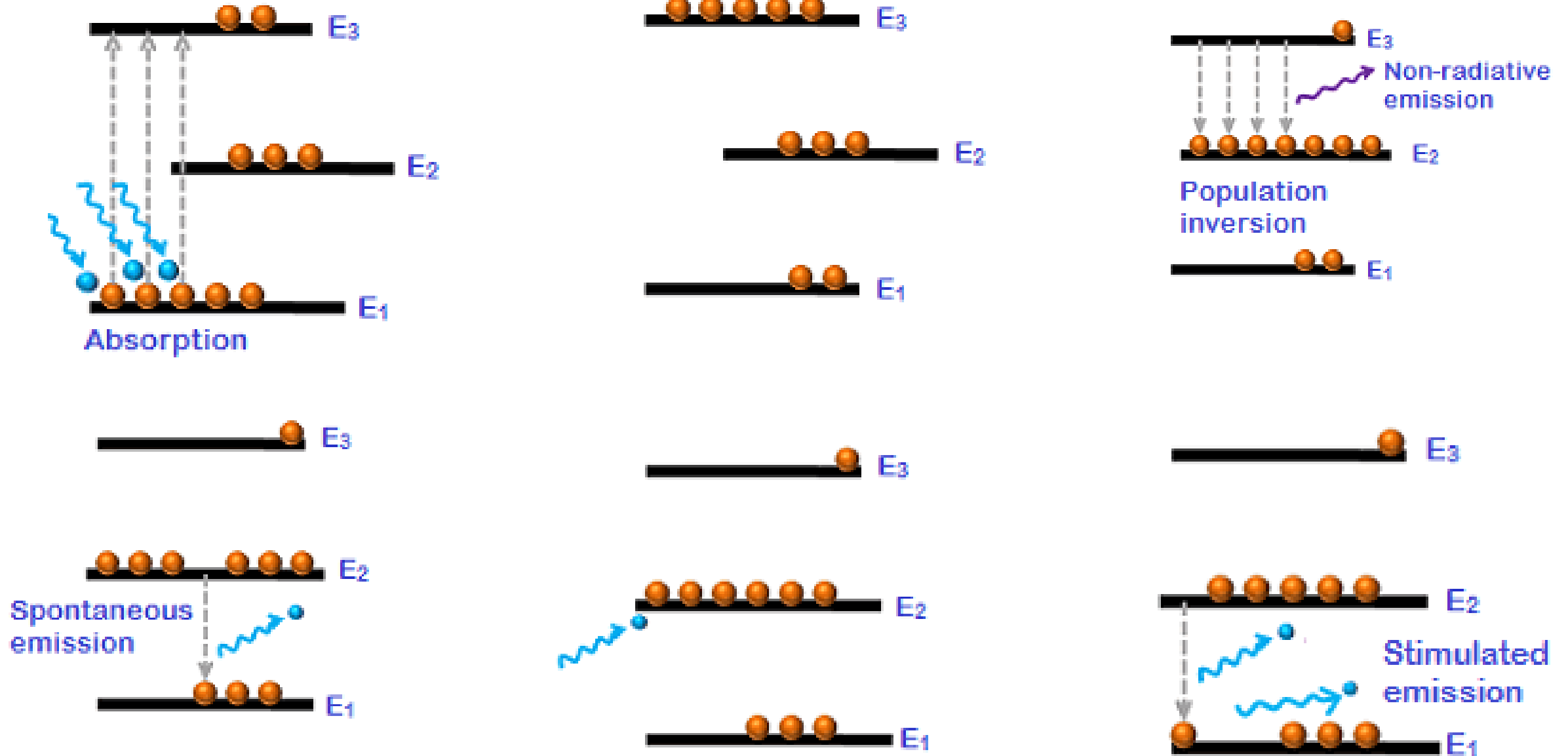


Four-level system



- Short-lived population inversion produce pulsed output – Ruby Laser
- Indefinite population inversion produce continuous output – termed CW (for continuous wave) lasers

Three-level system



Pumping – Types of Pumping Mechanisms

- Pumping – Process of raising - lower to higher energy levels – Population Inversion
- Optical Pumping - Ruby Laser
- Electrical Discharge – Helium Neon laser
- Direct conversion – Semiconductor Laser
- Chemical reaction – Carbondioxide laser

Basic Requirements – Production of a laser

1. Active Medium/ Active Material

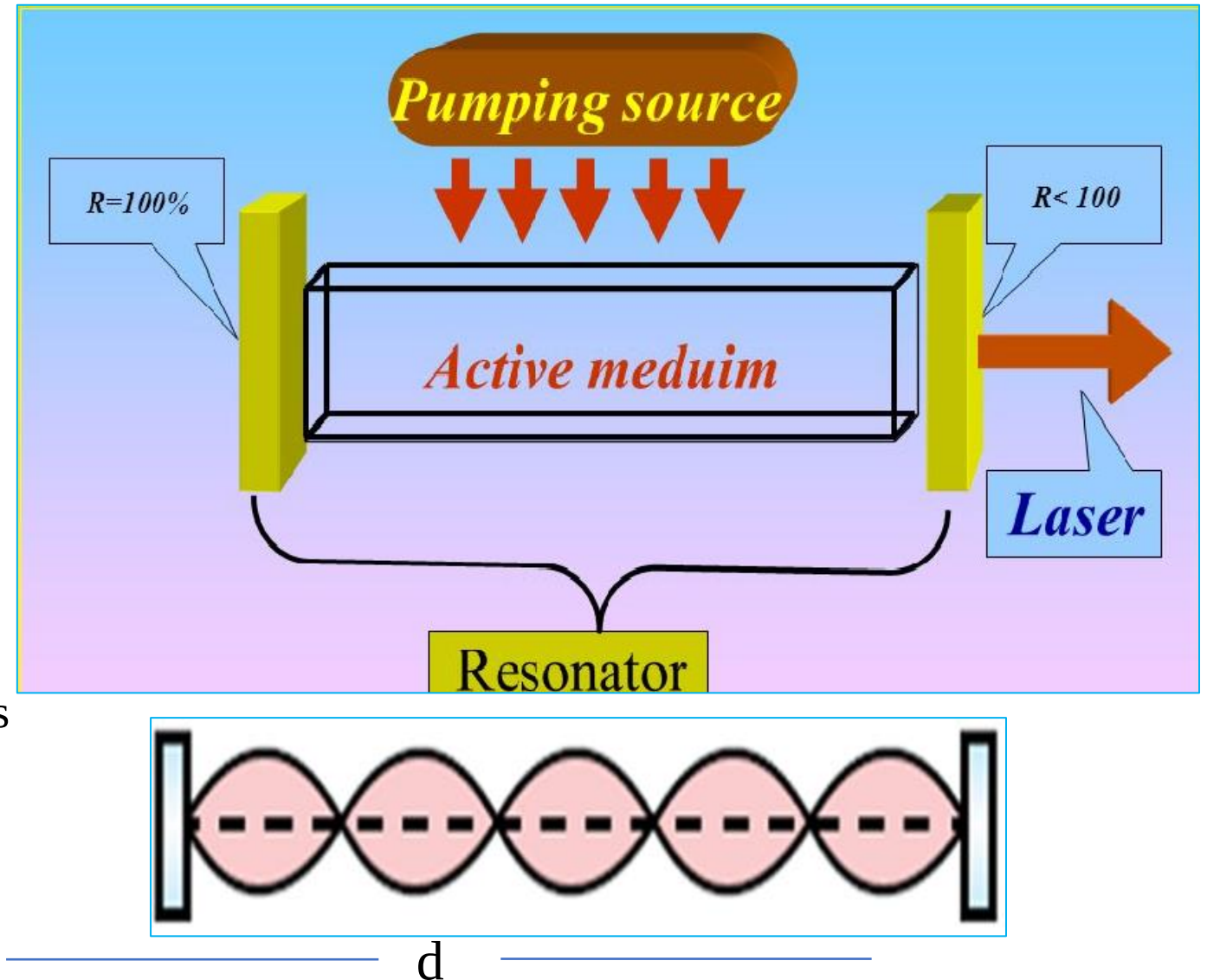
Atoms. Molecules or ions in solid, liquid or gaseous phase

2. Pumping Source

Necessary source that results in population Inversion. Active state

3. Optical Resonator/ Resonant Cavity

- Perfectly and partially reflecting mirrors that encloses active medium
- Optical Amplification – multiple reflections
- Mode – Standing wave (transverse Amplitude)
- Resonance condition : $\frac{2\pi}{\lambda}(2d) = m(2\pi)$
 d – the length of the cavity



Ruby laser/ Crystal Laser/ Three Level laser/ Pulsed Laser

Fabricated in 1960 by Theodore Maiman

Active Medium/ Active Material

- Ruby Crystal/ Chromium⁺³ ion
- Ruby – Crystal – Al_2O_3 , Cr_2O_3
- Ruby Rod – Pink in colour (0.5% of chromium)
- length – 4 cm, diameter - 0.5 cm

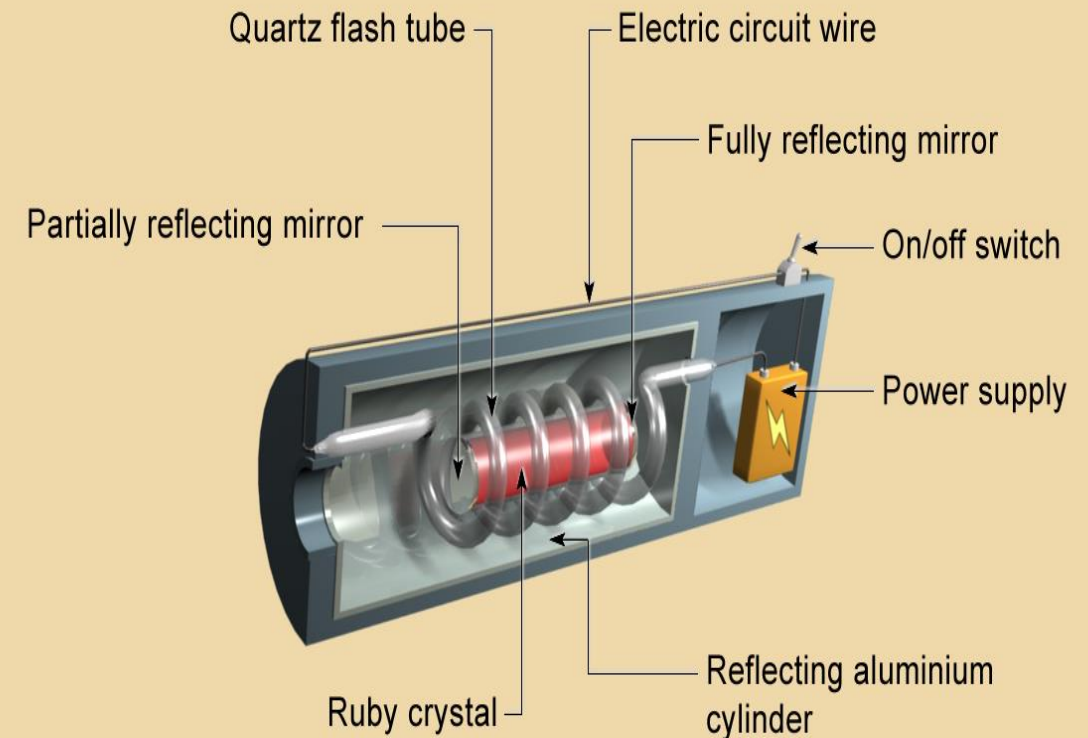
Pumping Source

- Quartz Flash tube (Xenon flash lamps)
- Green/ blue Component of white light

Optical Resonator/ Resonant Cavity

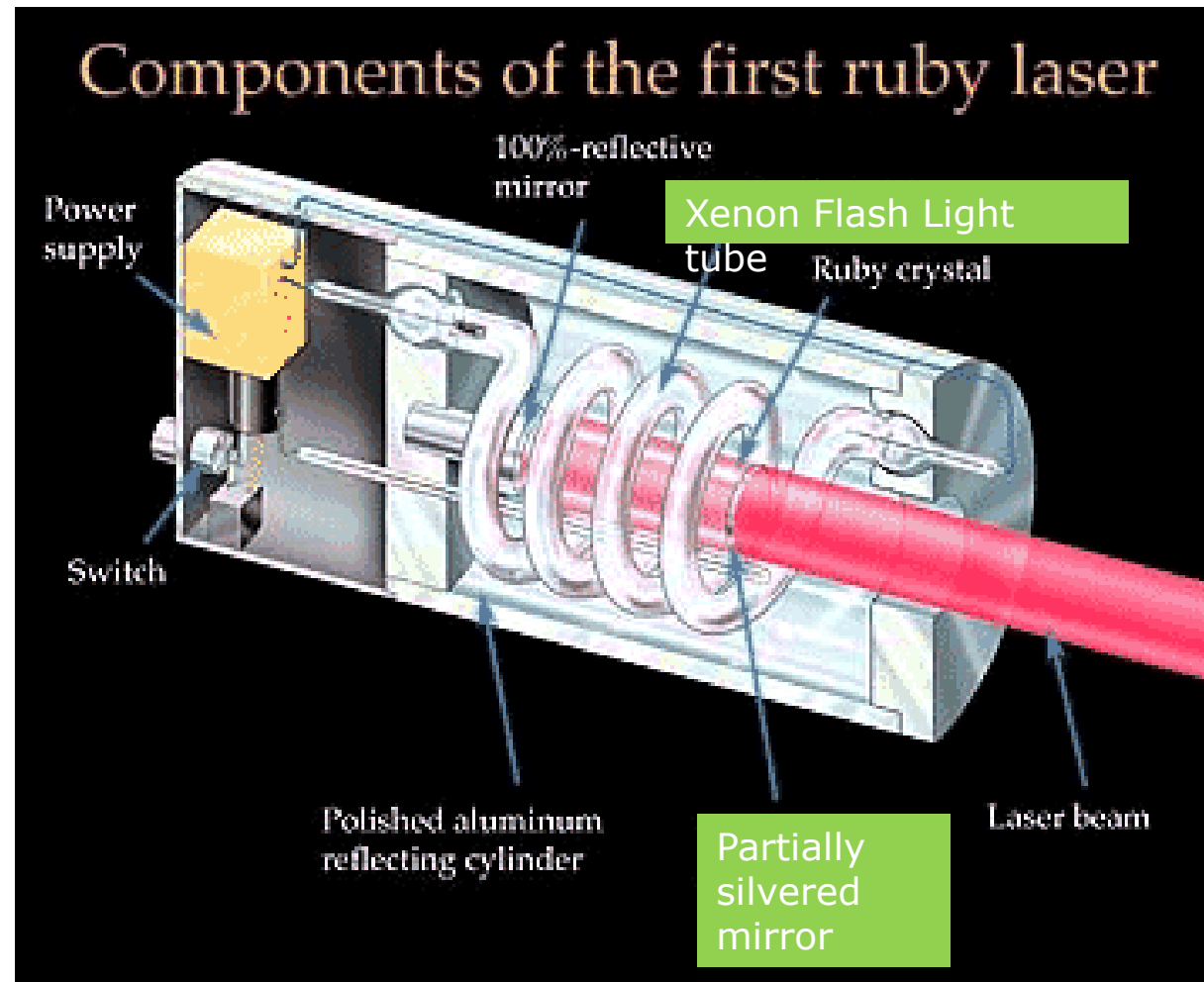
- Fully Reflecting – $R = 100\%$
- Partially Reflecting – $R < 100\%$
- Avalanche Effect

Cut-away View of a Ruby Laser

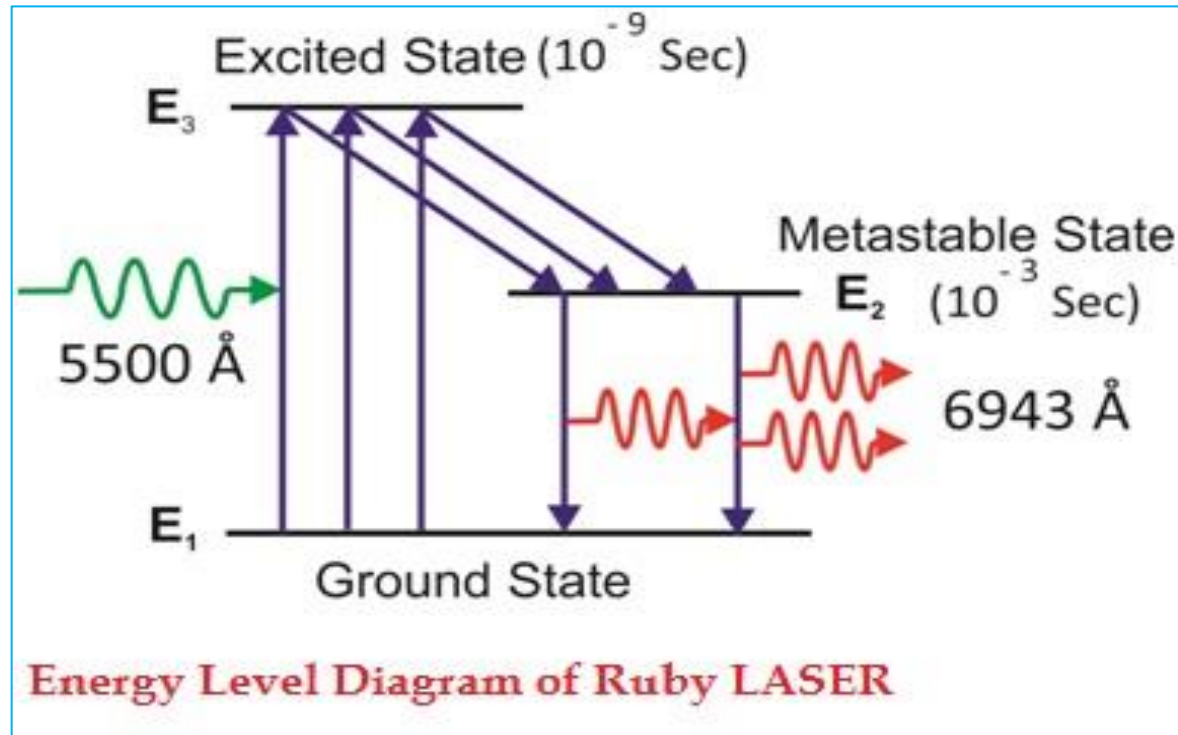


Ruby Laser (Three Level Laser)

Ruby (Al_2O_3) monocrystal, Cr doped.



Ruby laser – Energy Level Diagram

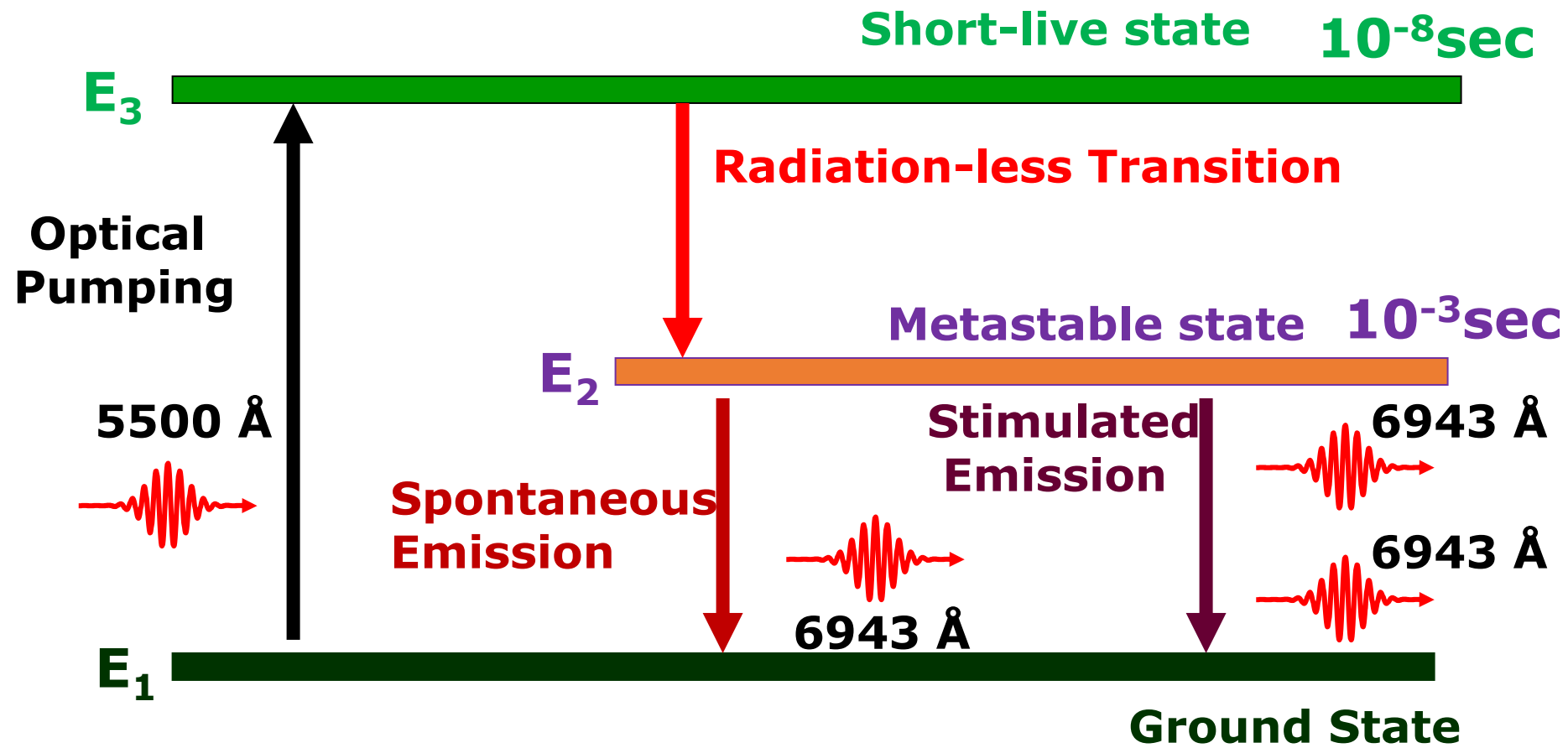


Drawbacks

$\lambda = 6943 \text{ \AA}$ or $6927 \text{ \AA}$

Intensity - 10000 Watt

1. Maximum pumping power
2. Discontinuous output
3. Low efficiency – crystal irregularities



Helium Neon laser/ Gas Laser/ Four Level laser/ Continuous Laser

1961 – Ali Jawan, William R. Bennett Jr, R. Heriott

Active Medium/ Active Material

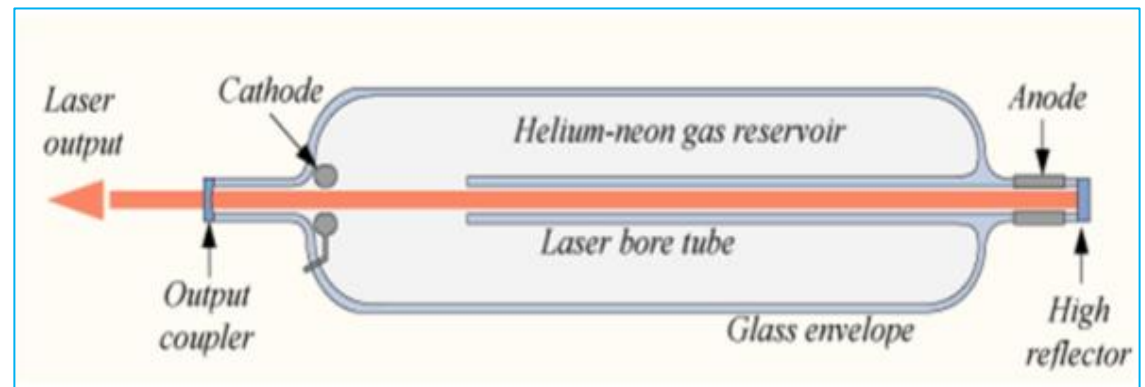
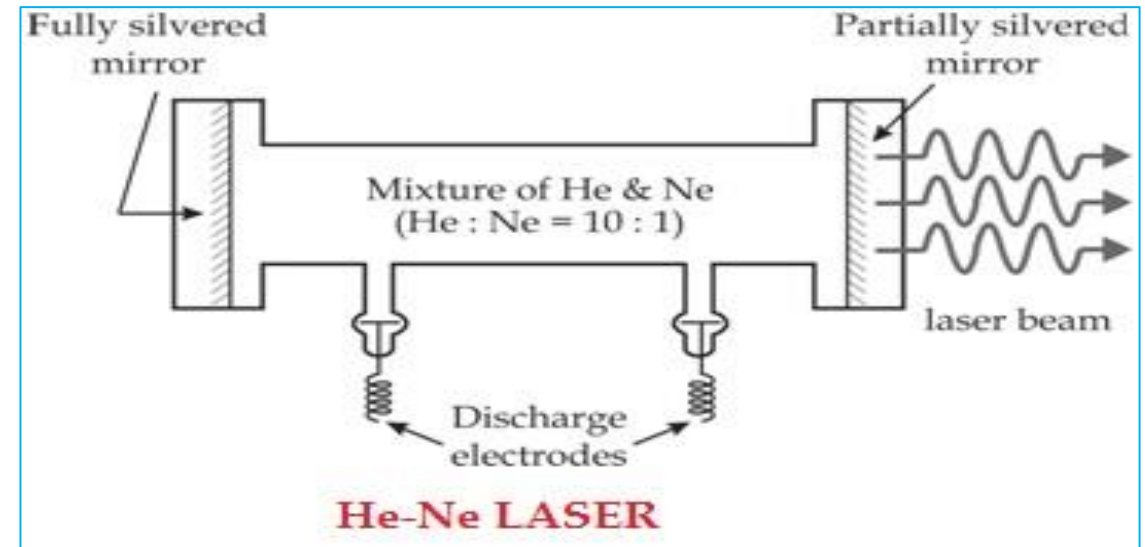
- Mixture of Helium Neon gases (10: 1)/ Neon
- Majority – Helium (1 mm of Hg), Minority – Neon (0.1 mm of Hg)
- length – 80 cm, diameter - 1.5 cm

Pumping Source

- Electrical discharge – RF Generator
- Operating Frequency – very high
- Input power = 50 Watt

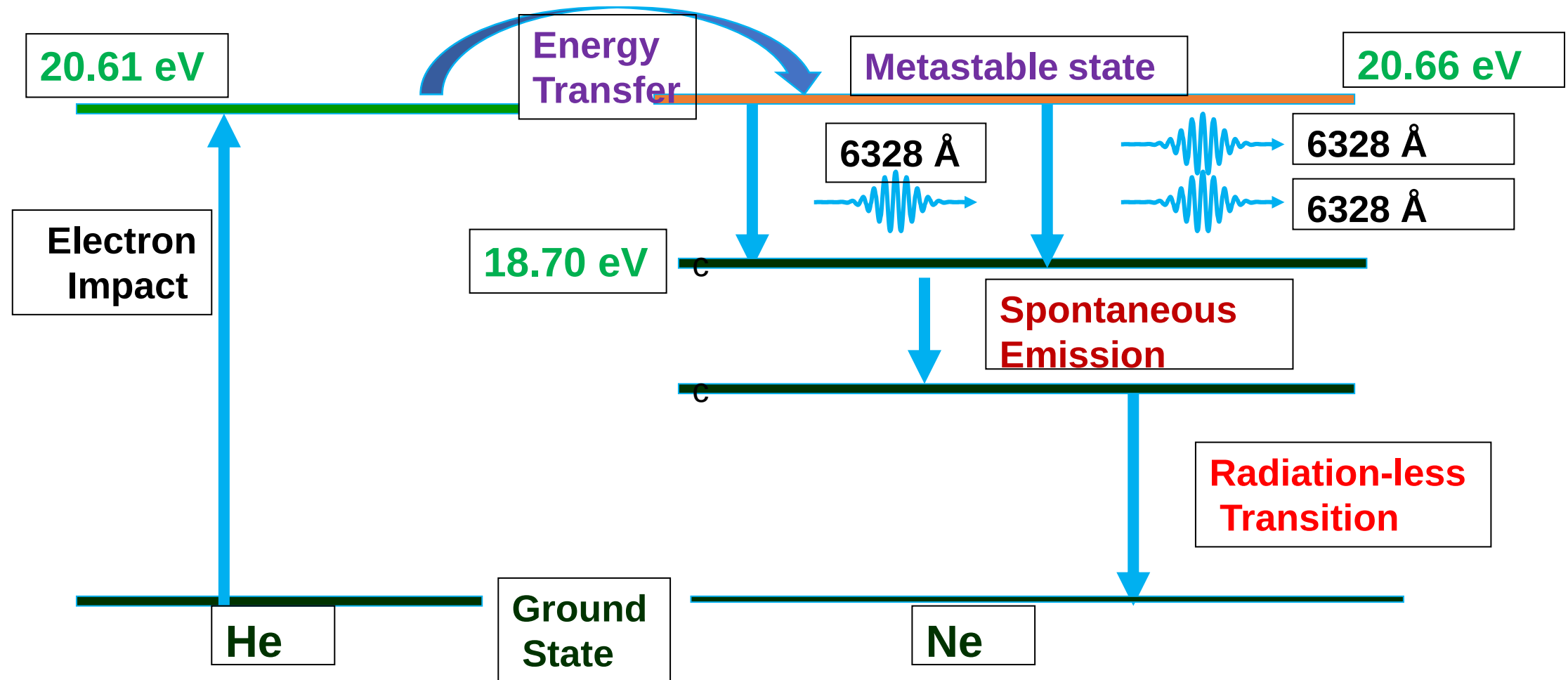
Optical Resonator/ Resonant Cavity

- Fully Reflecting – $R = 100\%$
- Partially Reflecting – $R < 100\%$
- Avalanche Effect



<https://emanzl.files.wordpress.com/2015/02/laser.png>

He- Ne laser – Energy Level Diagram



Ref: <https://emmanuelz.files.wordpress.com/2015/02/energy-level-diagra.png>

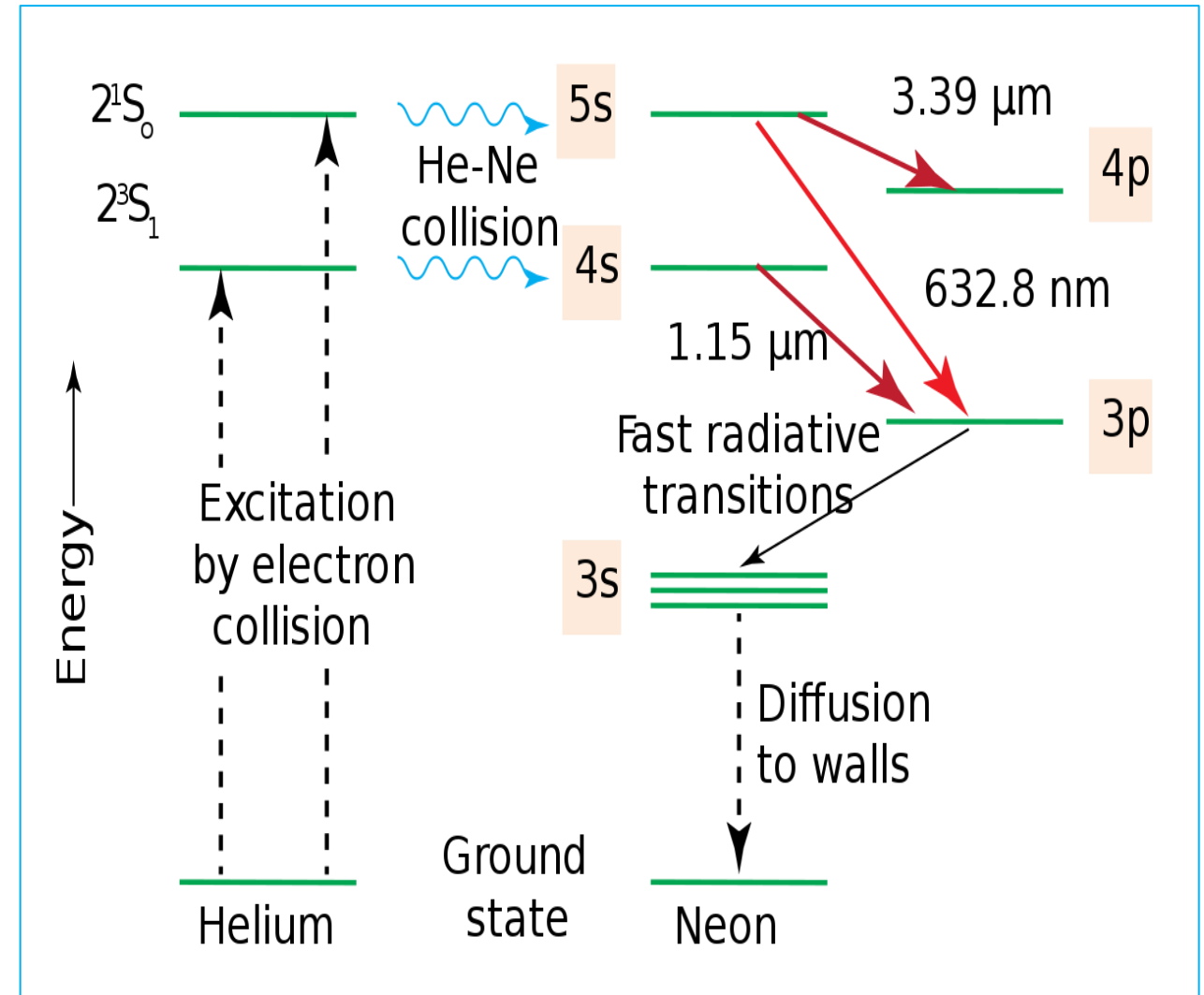
He- Ne laser – Energy Level Diagram

Advantages

- Highly Monochromatic ($\lambda = 6328 \text{ \AA}$)
- continuous output
- Pure spectrum and high stability of frequency

Disadvantages

- Moderate power



Photography vs Holography

Photography

- Incoherent light source, Normal sunlight
- Two dimensional image
- Records only the intensity variations
- Lenses are required
- One to one relation – object and image
- Image of the object to be photograph is recorded
- Negative is formed
- Photograph has less information capacity
- Irrecoverable loss if small portion of negative is destroyed
- Multiple images cannot be superimposed
- No parallax
- No Virtual Image

Holography

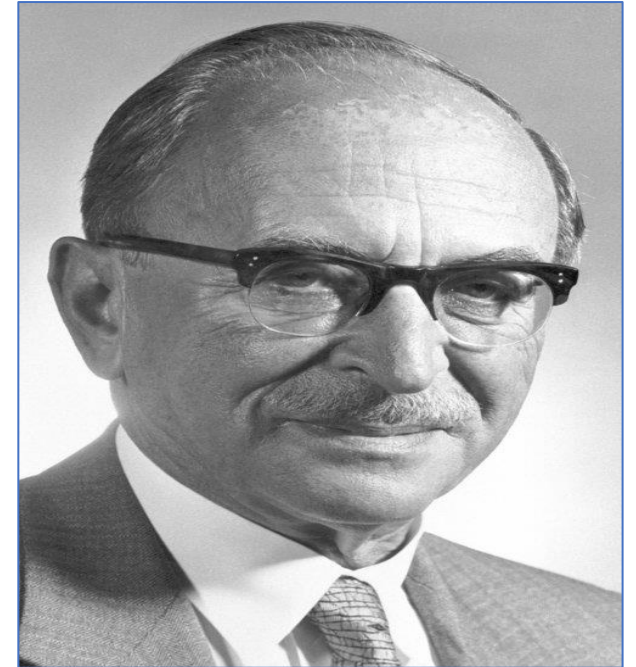
- High degree of coherence and monochromatic light (Lasers)
- Three dimensional image
- Records phase variations along with Intensity (Interferometric technique)
- Lensless photography
- No one to one relation- Object and image
- Image is not recorded but light waves reflected from object are recorded
- Hologram, a positive is formed
- Hologram has high information capacity
- Hologram into pieces, Image can be reconstructed with pieces of hologram
- Multiple images can be superimposed
- Exhibits parallax
- Virtual Image is formed

Holography

- Invented by Dennis Gabor at Research Lab of British Thomson Houston Company in 1947
- “Holo” – whole and “Graphy” – writing
- “Holography” – Complete Recording of an object
- Hologram – No resemblance with object, Information of object - Optical code
- Holography – Science of practicing making the Hologram
- Principle –Light Interference, Light Diffraction
- Two step Process – 1. Transformation/ Construction
2. Re-Transformation/ Re- Construction

Requirements

- Light waves of high degree of Coherence and Monochromatic light
- Source should emit narrow band instead of broad band – for good Resolution
- Path difference between object and reference waves shouldn't be greater than the coherence length
- Reference and scattered waves from the object interfere spatially
- Recording arrangement to be motionless
- Reconstruction source should have a constant wavelength and position



Dennis Gabor
(1900-1979)
Nobel prize: 1971
Invention: Holography
Father of Holography

Holography – Basic principle

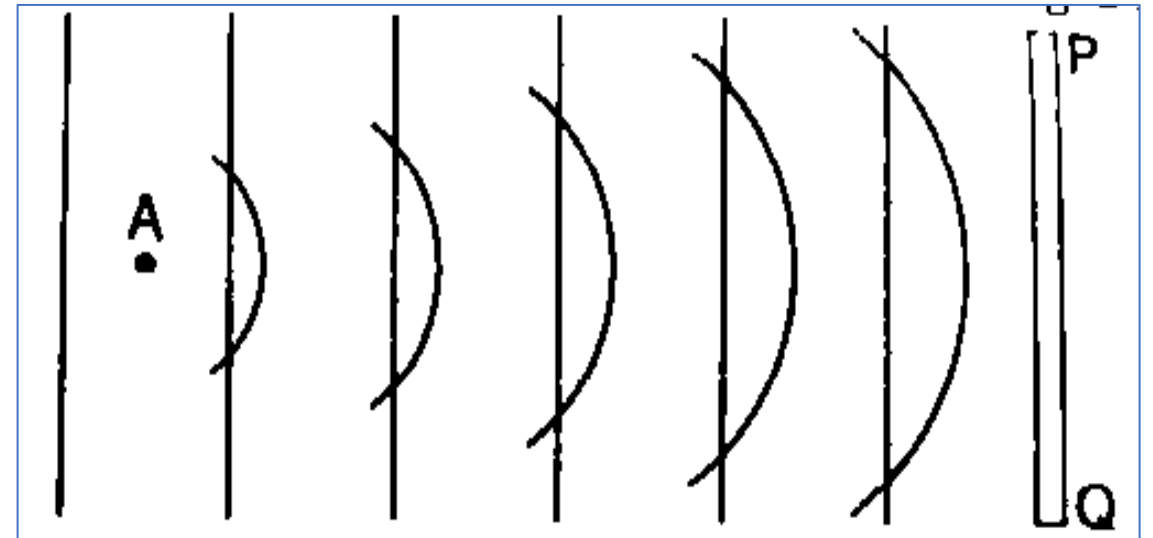
Point Object

Reference Wave – From the source – plane wave front

Object Wave- Coherent beam scattered (diffracted) from the object -spherical

Hologram – superposition of Reference wave and scattered wave.

Bright and dark concentric circles are formed



Solid

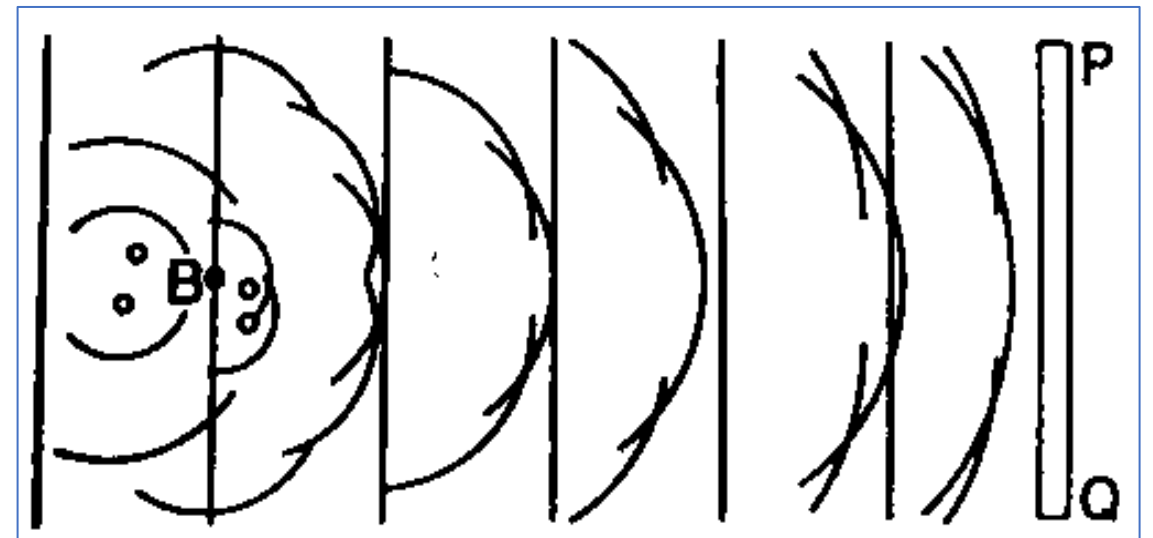
Collection of number of point objects

Reference Wave - From the source - plane wave front

Object Wave- Coherent beam scattered (diffracted) from each point

Hologram - superposition of Reference wave and scattered wave

Interference pattern – Complex – Intensity & Phase variations



Transformation/ Construction Process

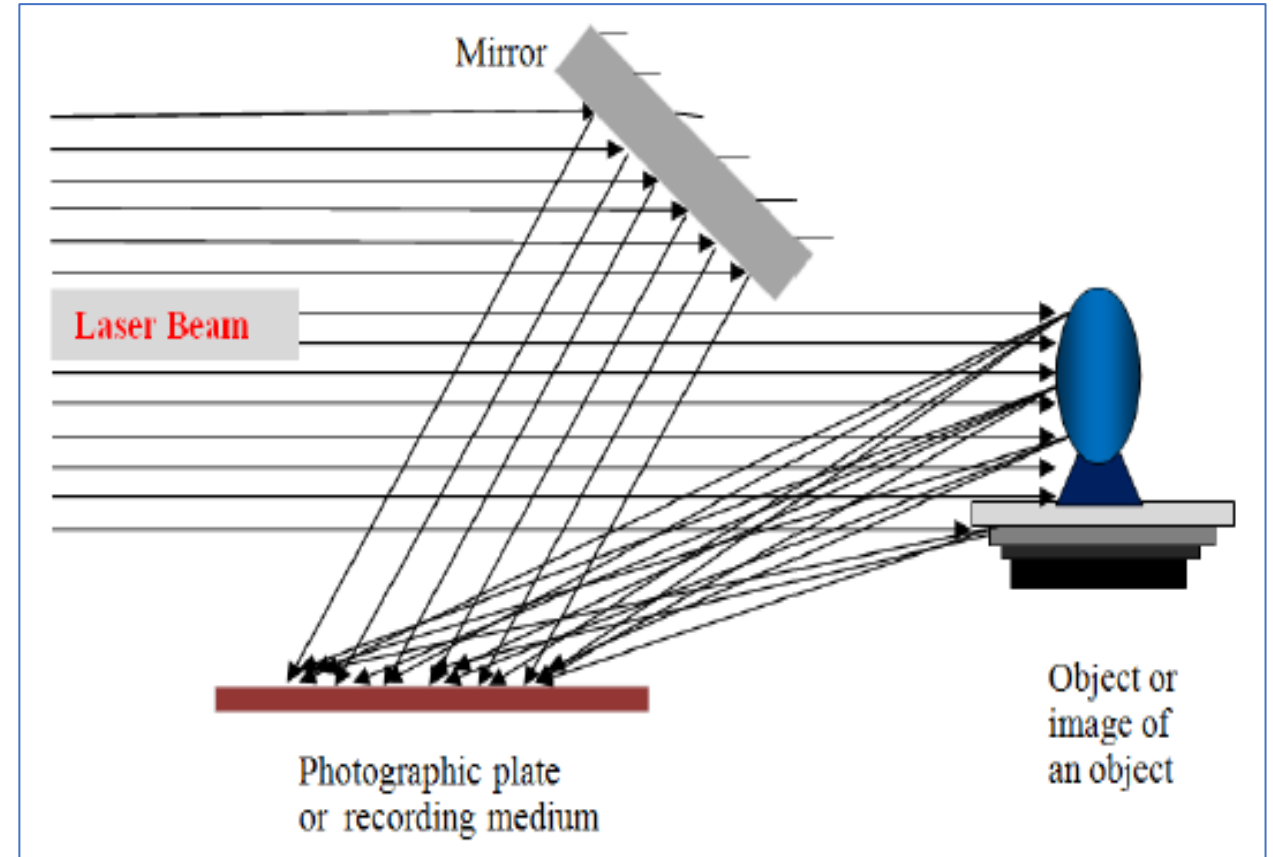
Laser Beam – Monochromatic and highly coherent

Beam Splitter – divides incident laser light into two parts

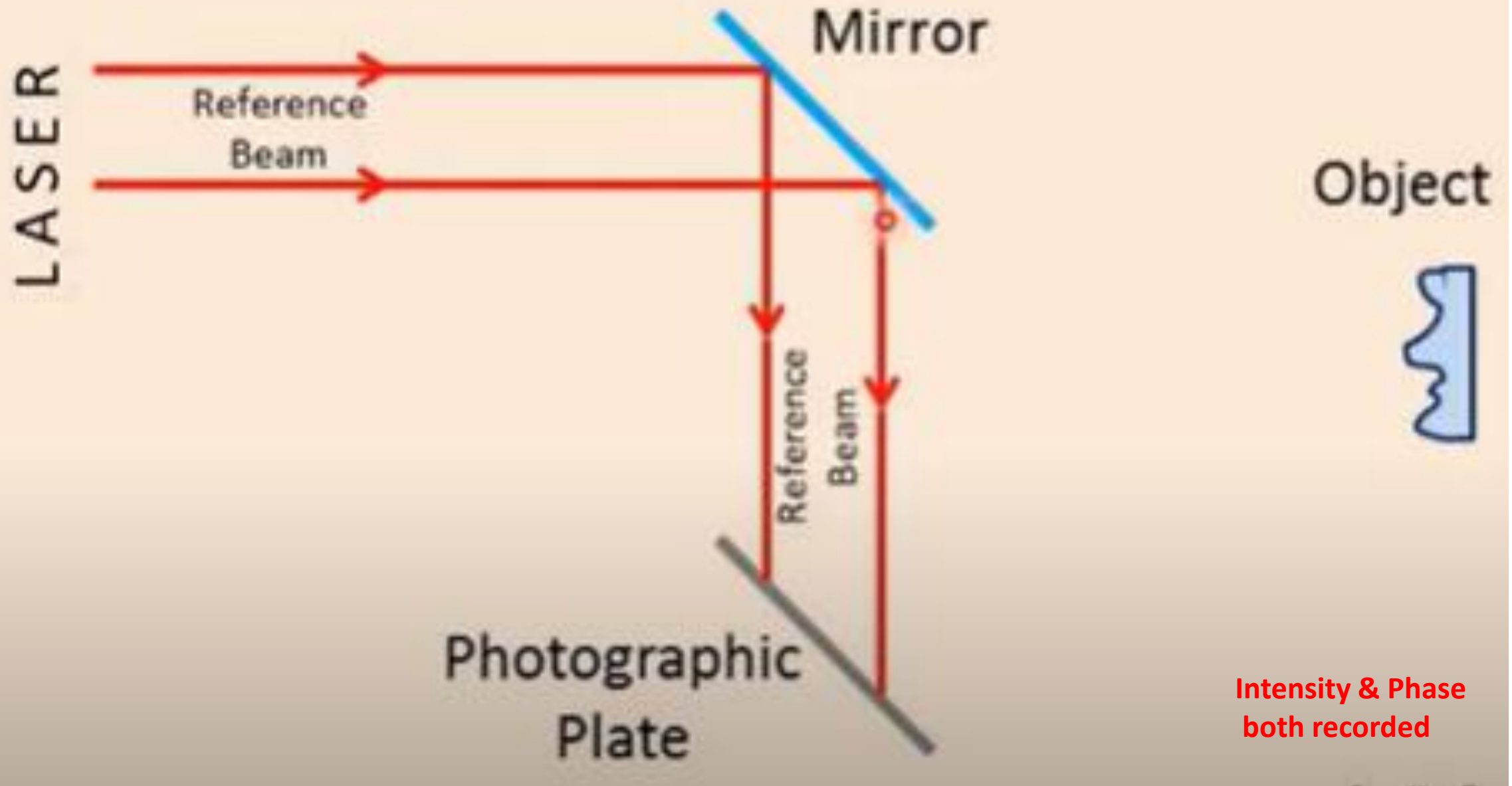
Plane wave Front – light rays reflected (Unobstructed – Ref Wave) from the mirror

Scattered wave front – Spherical - Amplitude and Phase

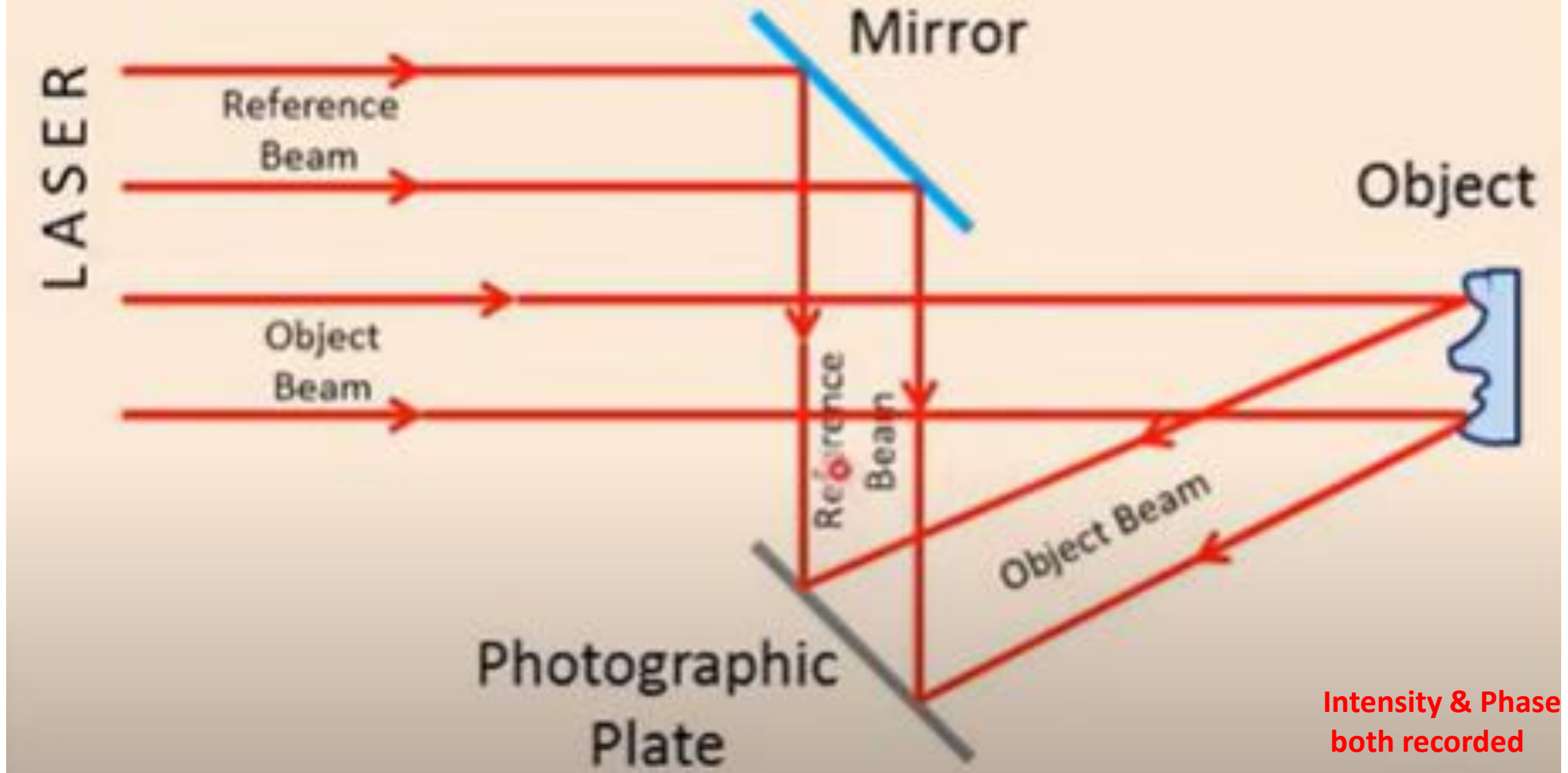
Complex Interference pattern – source code



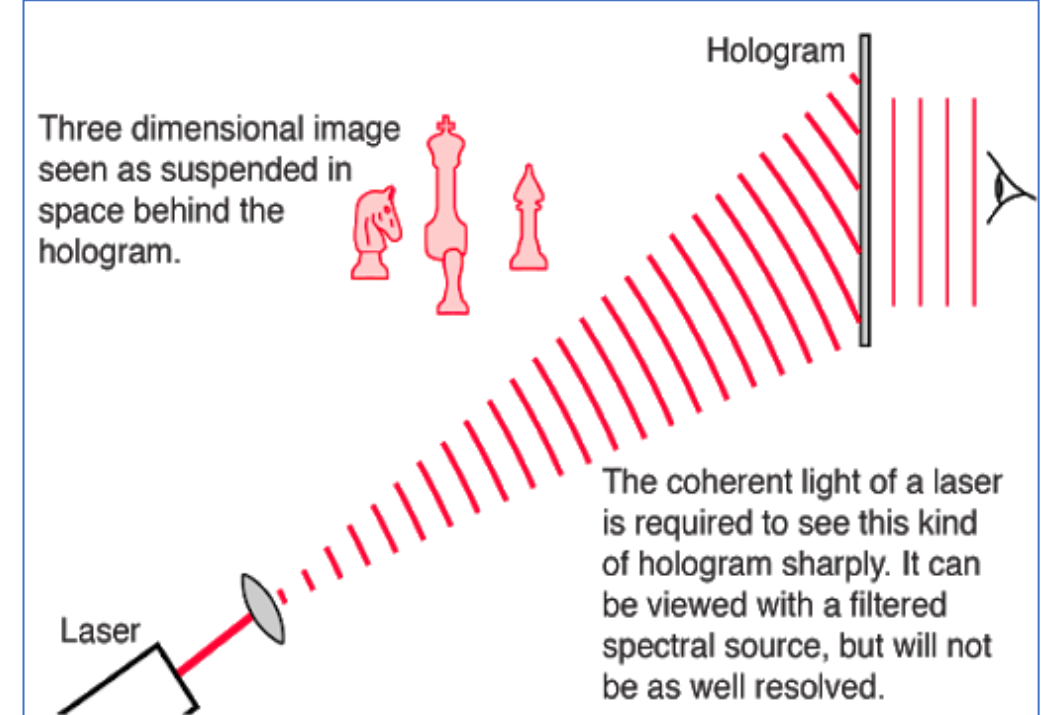
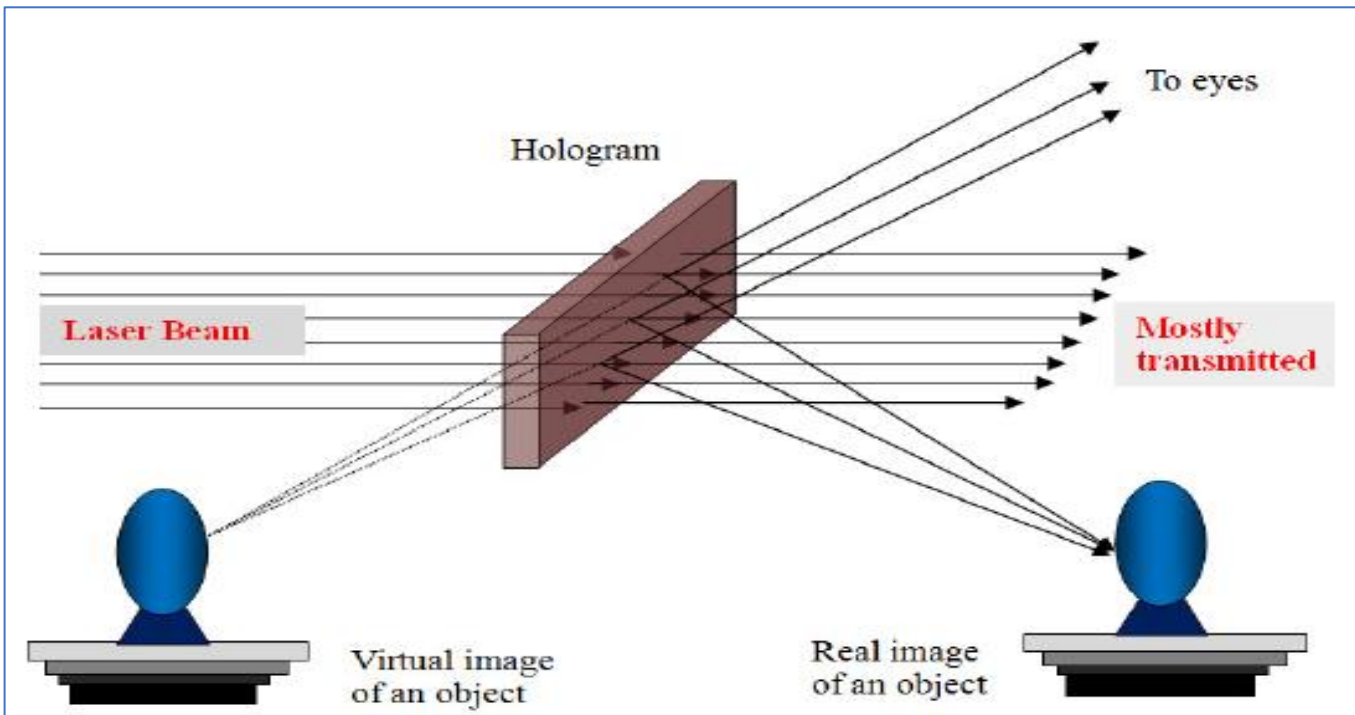
Construction of Hologram



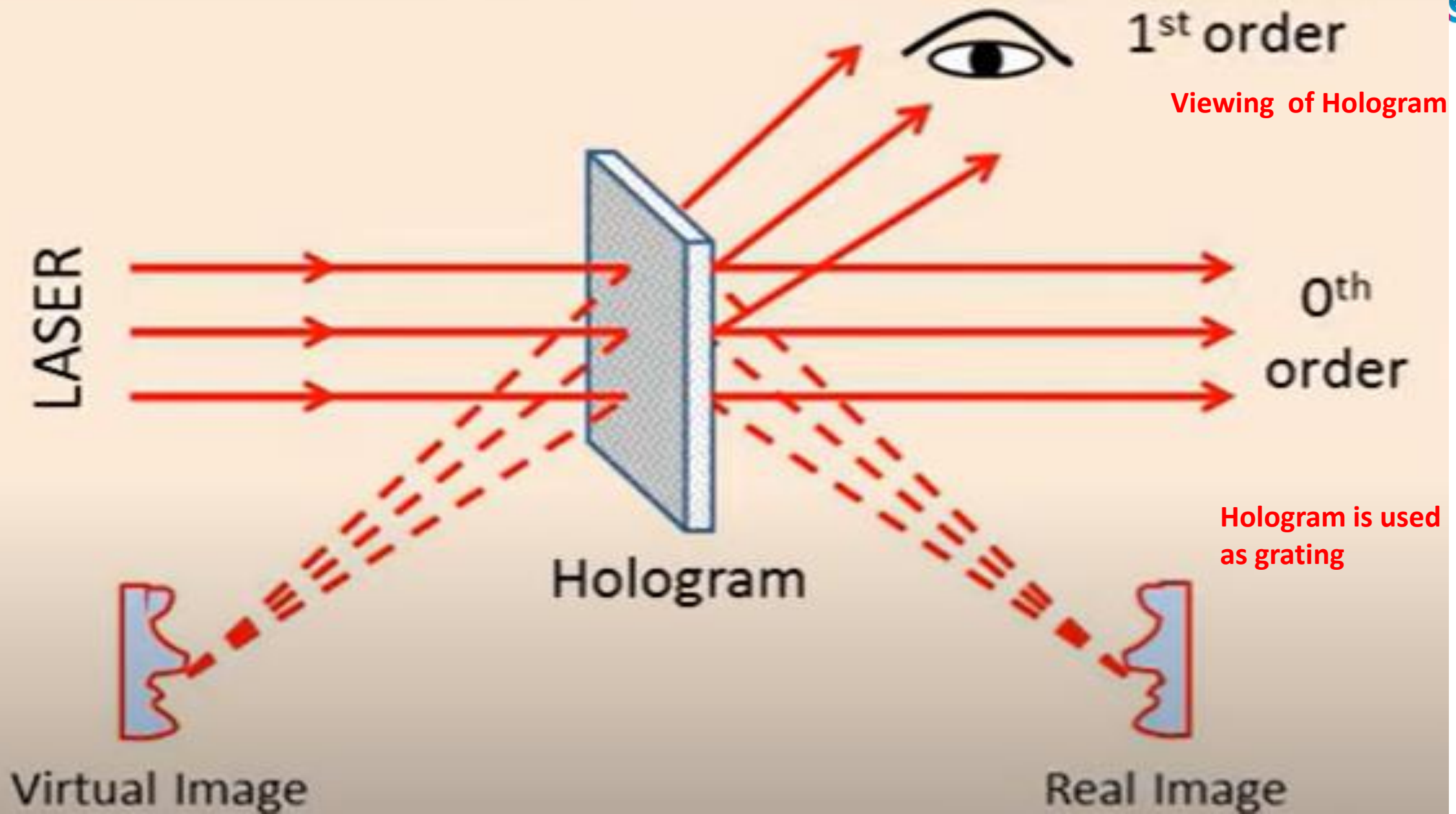
Construction of Hologram



Re-Transformation/ Re-Construction Process



- Same laser beam - construction
- Hologram – Complex Grating
- Zero Order – No information/ Amplitude variation only
- Amplitude and phase variations in all other directions
- Real Image – Photographed without aid of lens
- Virtual image – all the characteristics of the object





Thank You